LoanBazaar®
Secure Distribution Venture Pvt. Ltd.

# All About Corporate Finance Beginner to Expert ?

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## Executive Preface & Curriculum Architecture

Corporate finance forms the operational, strategic, and financial backbone of every business entity—from early-stage ventures to multi-billion-dollar multinational conglomerates. At its core, corporate finance is concerned with how corporations raise capital, allocate funds across competing investment opportunities, manage day-to-day liquidity, optimize capital structure, mitigate financial risks, and maximize long-term shareholder value.

This masterclass reference manual is engineered to serve as an all-inclusive, self-contained educational textbook. Reading this document alone is sufficient for any beginner, rookie, financial analyst, or corporate executive to master the end-to-end discipline of corporate finance. Every concept, mathematical formula, decision rule, accounting framework, and regulatory nuance is explained from first principles and supported by practical Indian corporate scenarios with currency figures denominated in Indian Rupees (INR / ₹).

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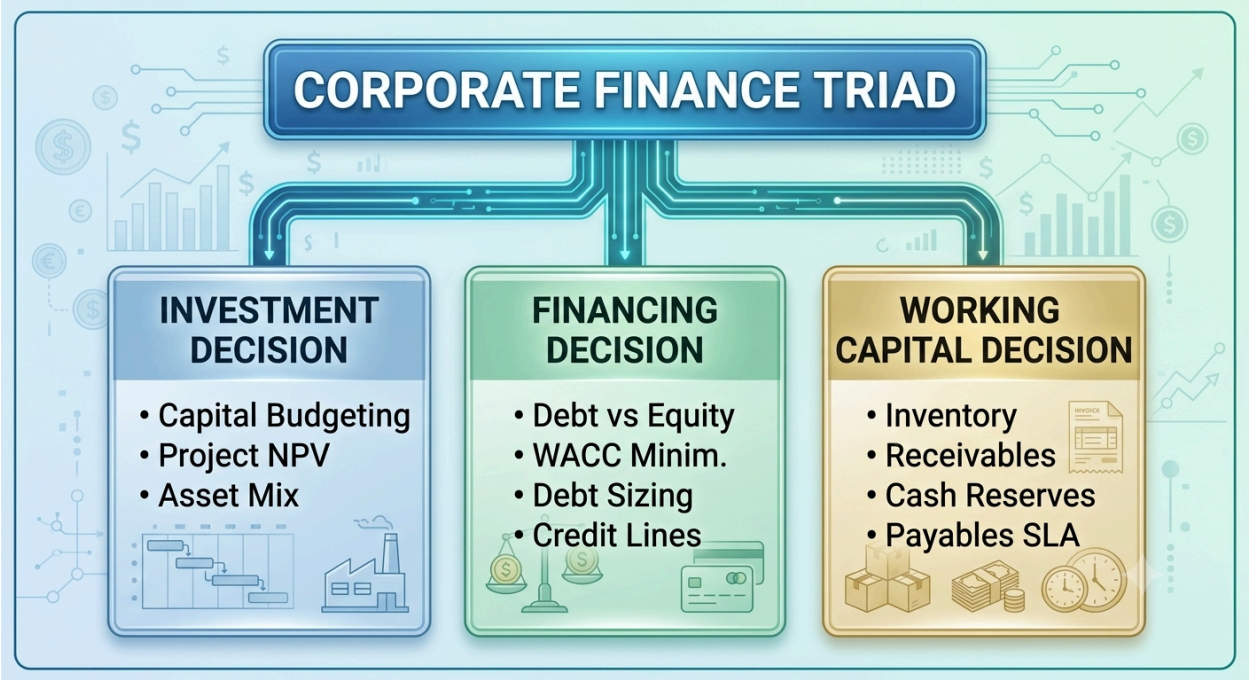
# Module 1: Fundamentals & Conceptual Framework of Corporate Finance

## 1.1 What is Corporate Finance?

Corporate finance is the area of finance dealing with the financial decisions corporations make and the tools and analysis used to make these decisions. The primary goal of corporate finance is to maximize corporate value while managing financial risks and balancing the interests of key stakeholders.

In modern enterprise management, corporate finance answers three fundamental questions: 1. **What long-term investments should the firm undertake?** (The Capital Budgeting / Investment Decision) 2. **How should the firm raise capital to fund these investments?** (The Financing / Capital Structure Decision) 3. **How should the firm manage its day-to-day operating cash flows and working capital?** (The Working Capital / Liquidity Decision)




## 1.2 Profit Maximization vs. Wealth Maximization

A foundational concept for beginners is understanding why **Shareholder Wealth Maximization** is superior to traditional **Profit Maximization** as the objective function of a corporation.

| Dimension | Profit Maximization | Wealth Maximization (NPV Maximization) |
|---|---|---|
| **Core Metric** | Short-term Net Accounting Profit (PAT). | Net Present Value (NPV) & Cash Flows. |

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| Dimension | Profit Maximization | Wealth Maximization (NPV Maximization) |
|---|---|---|
| **Time Horizon** | Short-sighted (focuses on annual accounting earnings). | Long-term (incorporates lifetime cash flows). |
| **Time Value of Money** | Ignores the Time Value of Money (treats ₹1 today equal to ₹1 in 5 years). | Fully incorporates TVM via Discounted Cash Flow (DCF). |
| **Risk & Uncertainty** | Ignores risk variability and cash flow quality. | Explicitly adjusts for risk via the discount rate (WACC / CAPM). |
| **Dividend Consideration** | Encourages retaining earnings even if return is low. | Optimizes cash payout if internal ROCE < Cost of Capital. |
| **Accounting Manipulations** | Vulnerable to accounting choices (depreciation, revenue timing). | Grounded in immutable cash inflows and outflows. |

## 1.3 Agency Theory & Corporate Governance

In modern corporations, ownership is separated from control. Shareholders (principals) own the firm, while managers (agents) control day-to-day operations. This separation creates the **Agency Problem**.

*Agency Conflicts in Corporate Finance:*

1. **Shareholder vs. Manager Conflicts**: Managers may prioritize personal prestige, executive perks, empire-building (unprofitable acquisitions), or risk aversion over shareholder returns.
2. **Shareholder vs. Debt Holder Conflicts**: Shareholders may engage in *asset substitution* (investing in ultra-risky projects with debt money) or *under-investment* (refusing to fund positive NPV projects when the benefits accrue primarily to bondholders).

*Agency Cost Mitigation Mechanisms:*

- **Performance-Linked Executive Compensation**: Stock options, Equity ESOPs, and bonus pools tied to Economic Value Added ($EVA$) and long-term stock performance.
- **Board Governance & Independent Directors**: SEBI (LODR) mandates requiring 50% independent directors and mandatory Audit Committee oversight.
- **Debt Covenants**: Affirmative and negative covenants imposed by lenders (e.g., maximum Debt/EBITDA ceilings, minimum DSCR requirements, restrictions on dividend payouts during default).
- **Threat of Hostile Takeovers & Capital Market Discipline**: Inefficient management risks acquisition and replacement by corporate raiders or private equity buyout funds.

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## 1.4 The Financial System & Capital Markets in India

A corporate finance manager operates within the broader financial infrastructure of India. Understanding the architecture of Indian financial markets is essential.

Regulatory Framework in India:

- **Reserve Bank of India (RBI)**: Regulates banking system liquidity, money markets, foreign exchange (FEMA), commercial paper, and External Commercial Borrowings (ECB).
- **Securities and Exchange Board of India (SEBI)**: Regulates equity capital markets, debt securities issuance, mutual funds, M&A takeover codes, and listed company disclosures.
- **Ministry of Corporate Affairs (MCA)**: Enforces the Companies Act, 2013, corporate governance rules, statutory filings (Form CHG-1/CHG-4), and financial reporting standards.
- **Insolvency and Bankruptcy Board of India (IBBI)**: Administers the Insolvency and Bankruptcy Code (IBC), 2016 for corporate debt resolution.

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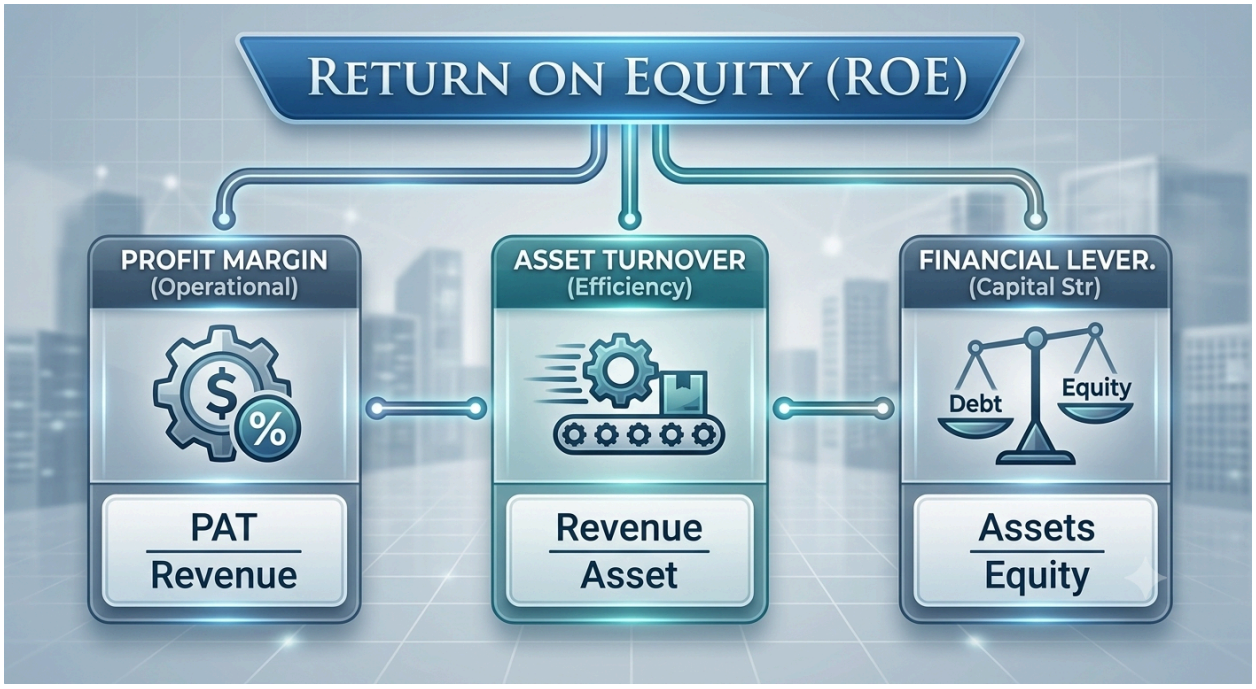


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# Module 2: Financial Statement Analysis & Accounting Foundations

## 2.1 The Four Core Financial Statements

Financial statement analysis is the diagnostic tool of corporate finance. Every financial manager must analyze four primary statements:

1. **Balance Sheet (Statement of Financial Position)**: A snapshot at a specific point in time showing Assets = Liabilities + Shareholders' Equity.
2. **Income Statement (Profit & Loss Account)**: A summary of revenues, expenses, and net profit over a financial period (e.g., FY 2025-26).
3. **Cash Flow Statement (CFS)**: Details the actual cash inflows and outflows categorized into Operating Activities (CFO), Investing Activities (CFI), and Financing Activities (CFF).
4. **Statement of Changes in Equity**: Tracks changes in share capital, retained earnings, and reserves.

## 2.2 Reconciling Accounting Profit to Cash Flow Metrics

A central tenet of corporate finance is that **cash flow pays the bills, not accounting profit**. Accountants use accrual accounting, whereas financial managers focus on cash generation.

*Key Profitability & Cash Flow Definitions:*

- **Earnings Before Interest, Tax, Depreciation & Amortization (EBITDA)**:

$$EBITDA = Revenue - COGS - Operating\ Expenses\ (excluding\ Dep\ \&Amort)$$

- **Earnings Before Interest & Tax (EBIT / Operating Profit)**:

$$EBIT = EBITDA - Depreciation - Amortization$$

- **Profit After Tax (PAT / Net Income)**:

$$PAT = (EBIT - Interest\ Expense) \times (1 - Tax\ Rate)$$

- **Operating Cash Flow (CFO)**:

$$CFO = PAT + Non - Cash\ Charges\ (Dep/Amort) \pm \Delta Working\ Capital$$

- **Free Cash Flow to Firm (FCFF)**:

$$FCFF = EBIT \times (1 - t) + Depreciation - Capex - \Delta Net\ Working\ Capital$$

- **Free Cash Flow to Equity (FCFE)**:

$$FCFE = FCFF - Interest \times (1 - t) + Net\ Debt\ Borrowed\ (Debt\ Issued - Debt\ Repaid)$$

## 2.3 Comprehensive Financial Ratio Analysis Framework

Financial ratios translate raw accounting numbers into actionable insights regarding liquidity, solvency, efficiency, profitability, and valuation.

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*A. Liquidity Ratios (Short-Term Solvency)*

- **Current Ratio**:

$$Current\ Ratio = \frac{Current\ Assets}{Current\ Liabilities} \quad (Target\ Benchmark:\ 1.25x - 1.50x)$$

- **Quick Ratio (Acid-Test Ratio)**:

$$Quick\ Ratio = \frac{Cash + Marketable\ Securities + Receivables}{Current\ Liabilities} \quad (Target\ Benchmark:\ \geq 1.00x)$$

- **Cash Ratio**:

$$Cash\ Ratio = \frac{Cash\ \&Bank\ Balances + Liquid\ Mutual\ Funds}{Current\ Liabilities}$$

*B. Solvency & Leverage Ratios (Long-Term Stability)*

- **Debt-to-Equity Ratio (D/E)**:

$$Debt - to - Equity = \frac{Total\ Interest-Bearing\ Debt}{Total\ Shareholders'\ Equity}$$

- **TOL / TNW Ratio (Total Outside Liabilities to Tangible Net Worth)**:

$$TOL\ /\ TNW = \frac{Total\ Outside\ Liabilities\ (Borrowings + Creditors + Provisions)}{Reported\ Equity - Intangible\ Assets - Accumulated\ Losses}$$

- **Interest Coverage Ratio (ICR)**:

$$ICR = \frac{EBIT}{Interest\ Expense} \quad (Investment\ Grade\ Benchmark:\ \geq 2.50x)$$

- **Debt Service Coverage Ratio (DSCR)**:

$$DSCR = \frac{PAT + Depreciation + Amortization + Interest}{Interest\ Expense + Scheduled\ Principal\ Repayments} \quad (Benchmark:\ \geq 1.25x - 1.35x)$$

*C. Operating Efficiency & Activity Ratios*

- **Inventory Holding Days (DIO)**:

$$DIO = \frac{Average\ Inventory}{Cost\ of\ Goods\ Sold\ (COGS)} \times 365$$

- **Debtors Collection Days (DSO)**:

$$DSO = \frac{Average\ Trade\ Receivables}{Gross\ Revenue\ from\ Operations} \times 365$$

- **Creditors Payment Days (DPO)**:

$$DPO = \frac{Average\ Trade\ Payables}{Cost\ of\ Goods\ Sold\ (or\ Cost\ of\ Raw\ Materials)} \times 365$$

- **Cash Conversion Cycle (CCC)**:

$$CCC = DIO + DSO - DPO$$

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*D. Profitability & Return Ratios*

- **Gross Profit Margin %**:

$$Gross\ Margin\ \% = \frac{Revenue - COGS}{Revenue} \times 100$$

- **EBITDA Margin %**:

$$EBITDA\ Margin\ \% = \frac{EBITDA}{Revenue} \times 100$$

- **Net Profit Margin %**:

$$Net\ Profit\ Margin\ \% = \frac{PAT}{Revenue} \times 100$$

- **Return on Capital Employed (ROCE %)**:

$$ROCE\ \% = \frac{EBIT}{Total\ Capital\ Employed\ (Total\ Assets - Current\ Liabilities)} \times 100$$

- **Return on Equity (ROE %)**:

$$ROE\ \% = \frac{PAT}{Total\ Shareholders'\ Equity} \times 100$$

## 2.4 The Du Pont Analysis Framework (3-Step & 5-Step)

Du Pont Analysis decomposes Return on Equity (ROE) to pinpoint the exact drivers of corporate performance.

*The 3-Step Du Pont Model:*

$$ROE = Net\ Profit\ Margin \times Asset\ Turnover \times Equity\ Multiplier$$

$$ROE = \left(\frac{PAT}{Revenue}\right) \times \left(\frac{Revenue}{Total\ Assets}\right) \times \left(\frac{Total\ Assets}{Equity}\right)$$

*The 5-Step Du Pont Model:*

$$ROE = Tax\ Burden \times Interest\ Burden \times Operating\ Margin \times Asset\ Turnover \times Leverage\ Multipl$$

$$ROE = \left(\frac{PAT}{EBT}\right) \times \left(\frac{EBT}{EBIT}\right) \times \left(\frac{EBIT}{Revenue}\right) \times \left(\frac{Revenue}{Total\ Assets}\right) \times \left(\frac{Total\ Assets}{Equity}\right)$$

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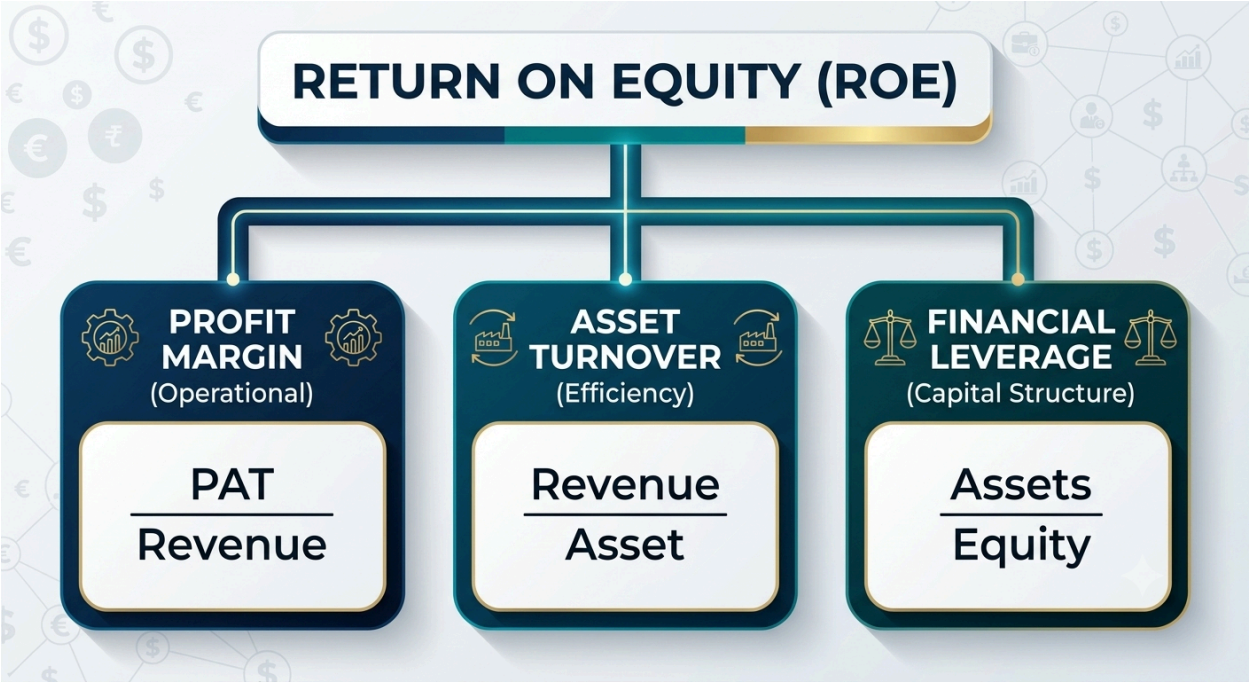


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# Module 3: Time Value of Money (TVM) & Mathematics of Finance

## 3.1 The Time Value of Money (TVM) Principle

The fundamental principle of corporate finance states that **a Rupee received today is worth more than a Rupee received in the future**. This is due to three reasons: 1. **Opportunity Cost / Interest Earning Potential**: Cash today can be invested to earn a return. 2. **Inflation**: Inflation erodes purchasing power over time. 3. **Risk & Uncertainty**: Future cash flows carry uncertainty and default risk.

## 3.2 Simple Interest vs. Compound Interest

- **Simple Interest**: Interest is earned only on the initial principal.

$$extFutureValue(Simple) = Pimes(1 + rimest)$$

- **Compound Interest**: Interest is earned on the principal plus accumulated interest from previous periods.

$$extFutureValue(Compound) = Pimes(1 + r)^{t}$$

*Effective Annual Rate (EAR) vs. Nominal Annual Percentage Rate (APR):*

When interest is compounded $m$ times per year:

$$extEAR = (1 + racextAPRmight)^{m} - 1$$

*Example*: An Indian bank quotes a working capital loan at a nominal interest rate of 12.0% p.a. compounded quarterly ($m = 4$).

$$extEAR = (1 + rac0.124ight)^{4} - 1 = (1.03)^{4} - 1 = 1.125508 - 1 = 12.55\%extp.a.$$

## 3.3 Present Value (PV) and Future Value (FV) Calculations

- **Present Value of a Single Future Cash Flow ($CF_t$)**:

$$PV = racCF_t(1 + r)^{t} = CF_times(1 + r)^{-t}$$

- **Future Value of a Present Lump Sum ($PV$)**:

$$FV_t = PVimes(1 + r)^{t}$$

## 3.4 Valuation of Annuities and Perpetuities

*A. Ordinary Annuity*

An ordinary annuity consists of a series of equal cash flows ($C$) occurring at the **end** of each period for $n$ periods. - **Present Value of Ordinary Annuity ($PV_{OA}$)**:

$$PV_{OA} = Cimes[rac1 - (1 + r)^{-n}right]$$

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- **Future Value of Ordinary Annuity ($FV_{OA}$)**:

$$FV_{OA} = Cimes[rac(1 + r)^{n} - 1right]$$

*B. Annuity Due*

An annuity due consists of cash flows occurring at the **beginning** of each period.

$$PV_{extAnnuityDue} = PV_{OA}imes(1 + r)$$

*C. Perpetuities (Infinite Stream of Cash Flows)*

- **Constant Perpetuity**: Cash flows ($C$) continue indefinitely at a constant discount rate ($r$).

$$PV_{extPerpetuity} = racCr$$

- **Growing Perpetuity**: Cash flows grow at a constant annual growth rate ($g$) where $r > g$.

$$PV_{extGrowingPerpetuity} = racC_{1}r - g$$

## 3.5 Amortization Schedule & Loan Amortization Mathematics

In corporate banking and term loan financing, loans are amortized via equal Equated Monthly Installments (EMI) or Equated Annual Installments.

*Equal Annual Installment Formula:*

$$extAnnualInstallment(A) = extPrincipal(P)imes[racrimes(1 + r)^{n}(1 + r)^{n} - 1ight]$$

*Numerical Amortization Example (INR):*

A corporate borrower takes a **₹10.00 Crore** equipment loan from an Indian commercial bank at an interest rate of **10.0% p.a.** payable in 5 equal annual installments.

$$A = ₹10,00,00,000imes[rac0.10imes(1.10)^{5}(1.10)^{5} - 1ight] = ₹10,00,00,000imes[rac0.1610510.$$

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# PROJECTED REPAYMENT SCHEDULE: ₹10 Cr TERM LOAN

(10% Rate, 5 Years)

## LOAN SUMMARY

- Loan Amount **₹10,00,00,000**
- Annual Rate **10%**
- Term **5 Years**
- Total Repayment **₹13,18,98,740**

| Year | Opening Principal (₹) | Installment (₹) | Interest @ 10% (₹) | Principal Repaid (₹) | Closing Principal (₹) |
|---|---|---|---|---|---|
| 1 | 10,00,00,000 | 2,63,79,748 | 1,00,00,000 | 1,63,79,748 | 8,36,20,252 |
| 2 | 8,36,20,252 | 2,63,79,748 | 83,62,025 | 1,80,17,723 | 6,56,02,529 |
| 3 | 6,56,02,529 | 2,63,79,748 | 65,60,253 | 1,98,19,495 | 4,57,83,034 |
| 4 | 4,57,83,034 | 2,63,79,748 | 45,78,303 | 2,18,01,445 | 2,39,81,589 |
| 5 | 2,39,81,589 | 2,63,79,748 | 23,98,159 | 2,39,81,589 | 0 |
| **TOTALS** | | Total Annual Installment (Outlay) **13,18,98,740** | Total Interest @ 10% **3,18,98,740** | Total Principal Repaid **10,00,00,000** | |

**PROJECT DEBT REPAYMENT SCHEDULE** | Fri, Sep 18, 2026 | Mumbai, Maharashtra, India

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# Module 4: Capital Budgeting & Investment Appraisal Techniques

## 4.1 The Capital Budgeting Decision Framework

Capital budgeting is the process through which a firm evaluates, sizes, and selects long-term capital investments that align with shareholder wealth maximization. Capital budgeting decisions are irreversible, involve massive cash outlays, and shape the firm's long-term competitive trajectory.

## 4.2 Incremental Cash Flow Estimation Rules

When evaluating a proposed project, corporate finance managers must analyze **incremental cash flows after tax (CFAT)**.

*Core Principles of Cash Flow Estimation:*

1. **Incremental Principle**: Measure cash flows *with* the project minus cash flows *without* the project.
2. **Post-Tax Basis**: All cash flows must be evaluated after deducting corporate taxes.
3. **Include Opportunity Costs**: If project uses existing land owned by the firm worth ₹10.0 Cr, that ₹10.0 Cr opportunity cost must be included as an initial outlay.
4. **Exclude Sunk Costs**: Past expenditures (e.g., ₹50.0 Lakhs spent last year on a market feasibility study) cannot be recovered and must be excluded.
5. **Include Side Effects (Cannibalization & Synergies)**: If a new product reduces sales of an existing product, that lost margin is an incremental cost.
6. **Include Working Capital Injections & Recoveries**: Initial inventory/receivables funding is an outflow at $t = 0$ and fully recovered as an inflow at project end ($t = n$).
7. **Depreciation Tax Shield**: Depreciation is a non-cash expense, but it reduces taxable income, creating a cash tax saving equal to $ ext{Depreciation} times t$.

## 4.3 Five Core Capital Budgeting Appraisal Techniques

*A. Net Present Value (NPV) — The Gold Standard*

NPV measures the net increase in shareholder wealth created by a project in present value terms.

$$extNPV = \sum_{t=1}^{n} racCF_{t}(1 + k)^{t} - C_{0}$$

- **Decision Rule**: Accept project if $ ext{NPV} > 0$. If evaluating mutually exclusive projects, choose the project with the highest positive NPV.

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*B. Internal Rate of Return (IRR)*

IRR is the discount rate ($r^*$) at which the present value of future cash inflows equals the initial investment outlay ($ ext{NPV} = 0$).

$$0 = \sum_{t=1}^{n} racCF_t(1 + extIRR)^t - C_0$$

- **Decision Rule**: Accept project if $ ext{IRR} > ext{Cost of Capital (WACC)}$.

*C. Modified Internal Rate of Return (MIRR)*

MIRR addresses the main flaw of standard IRR (which unrealistically assumes intermediate cash flows are reinvested at the IRR itself). MIRR assumes intermediate cash inflows are reinvested at the firm's WACC.

$$extMIRR = (racextTerminalValueofInflowsatWACCextPresentValueofOutlaysatWACCight)^{rac1n} -$$

*D. Profitability Index (PI) / Benefit-Cost Ratio*

$$extProfitabilityIndex(PI) = racextPresentValueofFutureCashInflowsextInitialInvestmentOutlay($$

- **Decision Rule**: Accept if $ ext{PI} > 1.0$. Excellent for ranking independent projects under **Capital Rationing**.

*E. Payback Period (PBP) & Discounted Payback Period (DPBP)*

- **Payback Period**: Time required for cumulative nominal cash inflows to recover the initial investment.
- **Discounted Payback Period**: Time required for cumulative *discounted* cash inflows to recover the initial outlay.

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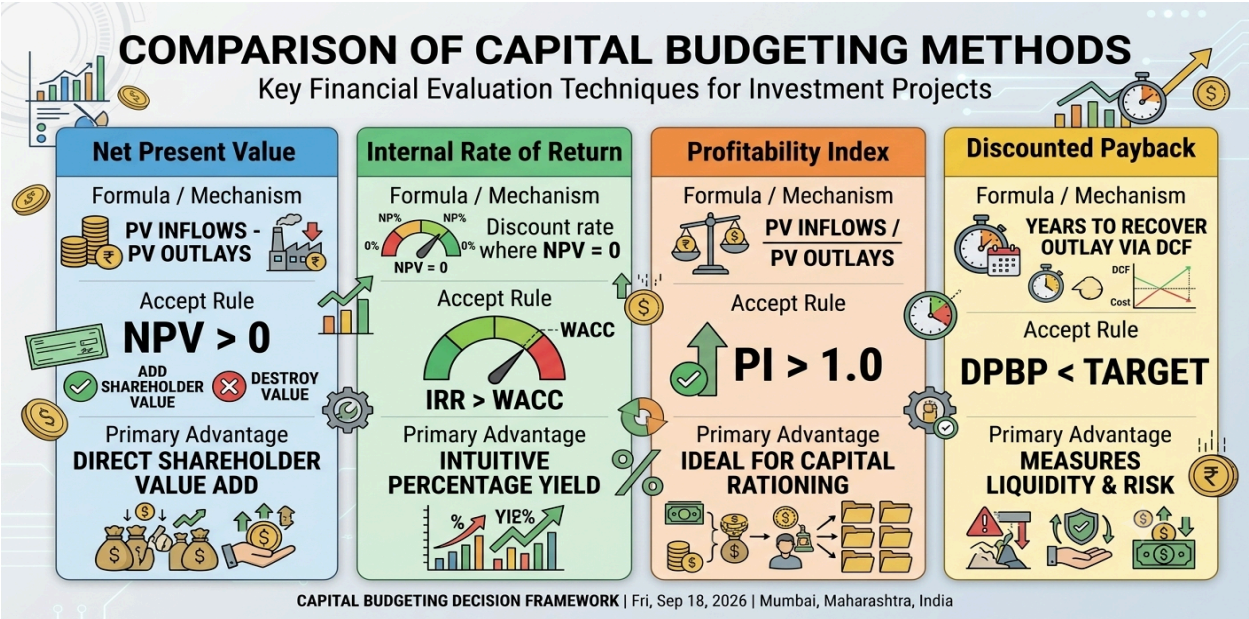
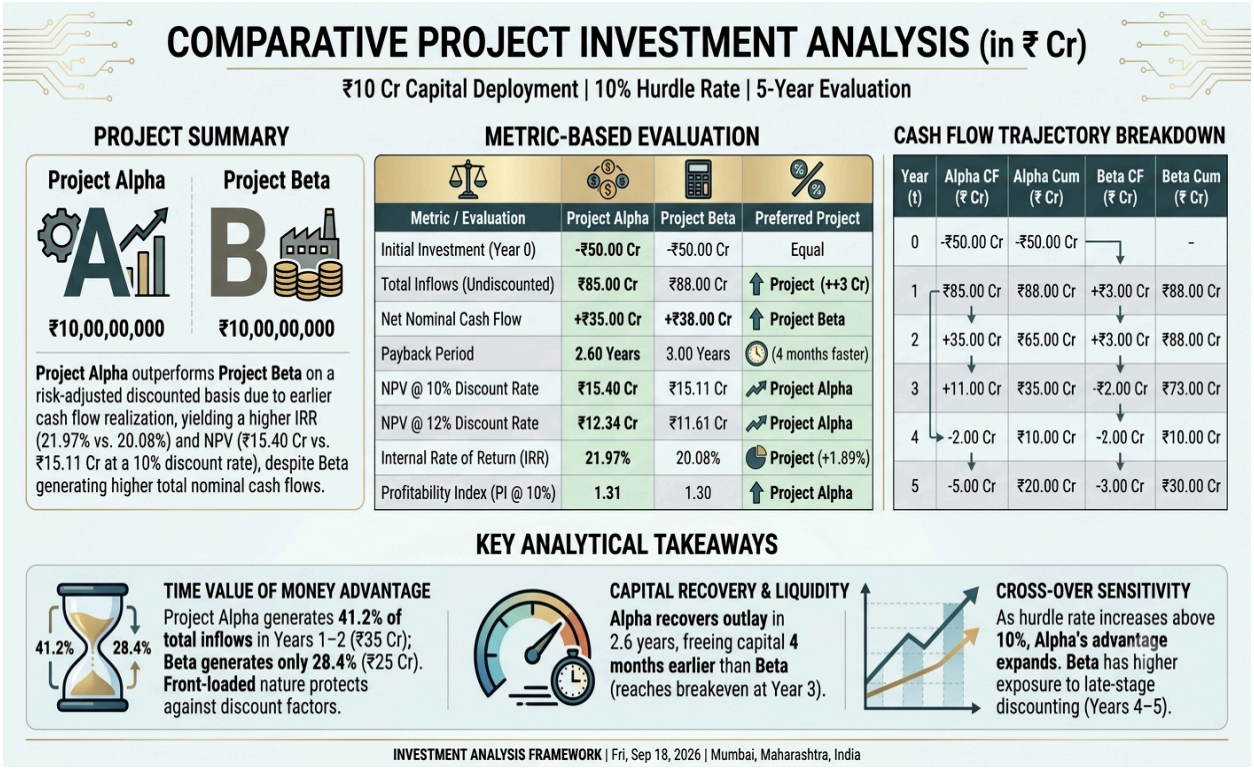



## 4.4 Comprehensive Capital Budgeting Comparison Illustration (INR)

An Indian industrial manufacturer (*Gujarat Precision Components Ltd.*) is evaluating two mutually exclusive equipment expansion projects: **Project Alpha (Automated Machining)** and **Project Beta (Robotic Assembly)**. Both projects require an initial outlay of **₹50.00 Crores** at $t = 0$. The company's WACC is **10.0% p.a.**

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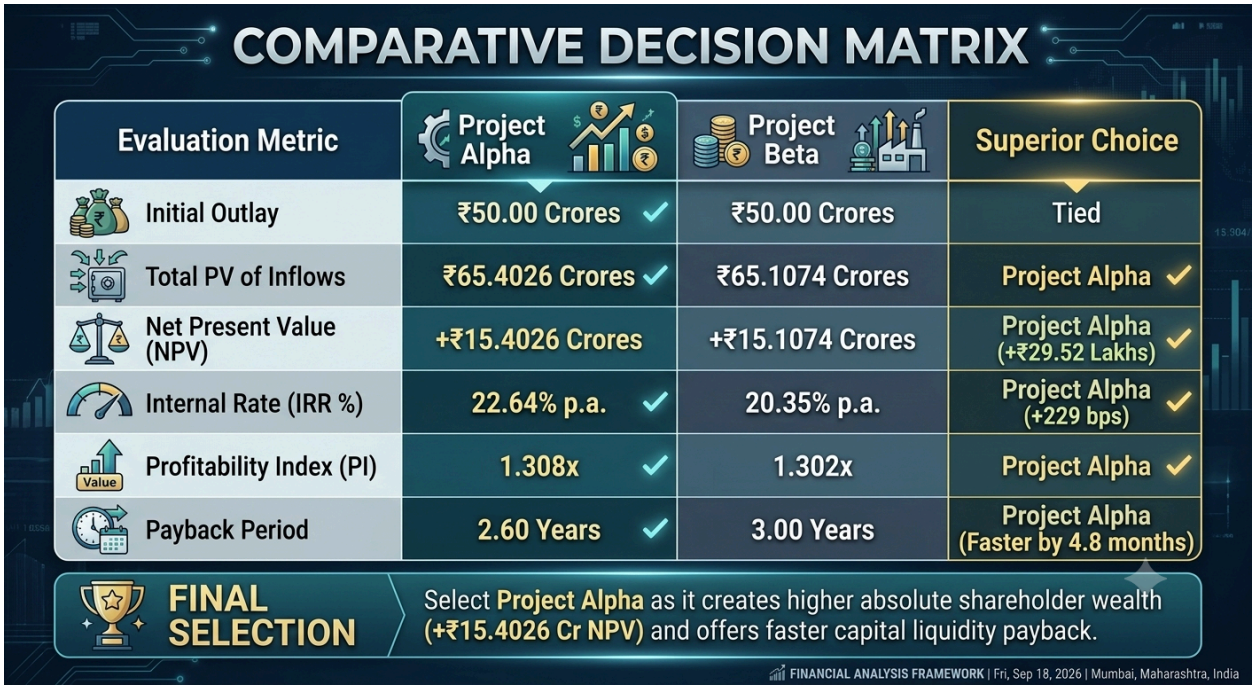
*Step-by-Step Mathematical Evaluation:*

**1. Project Alpha Calculations (WACC = 10%):** - $PV_1 = 15.00/(1.10)^1 = 13.6364$ - $PV_2 = 20.00/(1.10)^2 = 16.5289$ - $PV_3 = 25.00/(1.10)^3 = 18.7829$ - $PV_4 = 15.00/(1.10)^4 = 10.2452$ - $PV_5 = 10.00/(1.10)^5 = 6.2092$ - **Total PV of Inflows (Alpha)** $= 13.6364 + 16.5289 + 18.7829 + 10.2452 + 6.2092 = ₹65.4026 extCrores$ - **NPV (Alpha)** $= 65.4026 - 50.00 =+ ₹15.4026 extCrores$ - **Profitability Index (Alpha)** $= 65.4026/50.00 = 1.308x$ - **IRR (Alpha)** $= 22.64\% extp.a.$ - **Payback Period (Alpha)**: Year 1 (15) + Year 2 (20) = 35. Need 15 in Year 3 $ ightarrow 2 + (15/25) = $.

**2. Project Beta Calculations (WACC = 10%):** - $PV_1 = 10.00/(1.10)^1 = 9.0909$ - $PV_2 = 15.00/(1.10)^2 = 12.3967$ - $PV_3 = 25.00/(1.10)^3 = 18.7829$ - $PV_4 = 20.00/(1.10)^4 = 13.6603$ - $PV_5 = 18.00/(1.10)^5 = 11.1766$ - **Total PV of Inflows (Beta)** $= 9.0909 + 12.3967 + 18.7829 + 13.6603 + 11.1766 = ₹65.1074 extCrores$ - **NPV (Beta)** $= 65.1074 - 50.00 =+ ₹15.1074 extCrores$ - **Profitability Index (Beta)** $= 65.1074/50.00 = 1.302x$ - **IRR (Beta)** $= 20.35\% extp.a.$ - **Payback Period (Beta)**: Year 1 (10) + Year 2 (15) + Year 3 (25) = 50. Exactly $3.00 extYears$.




## 4.5 Risk Analysis in Capital Budgeting

In real-world corporate finance, future cash flows are uncertain. Three analytical tools are used to incorporate risk:

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1. **Sensitivity Analysis**: Changes one variable at a time (e.g., sales volume, raw material cost, discount rate) to measure its impact on NPV.
2. **Scenario Analysis**: Evaluates discrete scenarios (*Base Case, Optimistic Case, Pessimistic / Bear Case*) where multiple parameters change simultaneously.
3. **Real Options Analysis**: Recognizes that corporate managers possess managerial flexibility to modify decisions over the project lifecycle:
   - *Option to Expand*: Scaling up production if market demand exceeds expectations.
   - *Option to Abandon*: Terminating a project early and selling assets for salvage value if performance drops.
   - *Option to Delay / Defer*: Waiting for regulatory or market clarity before committing capital.

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# Module 5: Cost of Capital & Weighted Average Cost of Capital (WACC)

## 5.1 Concept and Significance of Cost of Capital

The **Cost of Capital** is the minimum required rate of return that a corporation must earn on its investments to satisfy its capital providers (equity shareholders, debt holders, and preference shareholders) and maintain its enterprise market value. It serves two critical functions in corporate finance: 1. **The Hurdle Rate**: Used as the discount rate in Capital Budgeting (NPV & IRR). 2. **The Benchmark for Capital Structure Optimization**: The optimal capital structure is defined as the debt-equity mix that **minimizes WACC** and **maximizes firm value**.

## 5.2 Component Cost of Capital Calculations

*A. Cost of Debt ($K_d$)*

Debt carries explicit interest payments. Because interest expense is tax-deductible under Section 36(1)(iii) of the Income Tax Act, 1961 in India, the effective cost of debt to a tax-paying corporation is significantly lower than the gross coupon rate.

- **Pre-Tax Cost of Debt ($K_{d,pre}$)**: The yield to maturity (YTM) or interest rate on corporate debt.
- **Post-Tax Cost of Debt ($K_{d,post}$)**:

$$K_d = K_{d,pre} \times (1 - t)$$

Where $t$ is the marginal corporate tax rate (including surcharge & cess, e.g., 25.168% under Section 115BAA).

*Example*: An Indian firm issues 10.50% p.a. corporate debentures. At a corporate tax rate of 25.168%:

$$K_d = 10.50\% \times (1 - 0.25168) = 10.50\% \times 0.74832 = 7.857\%\, p.a.$$

*B. Cost of Preference Shares ($K_p$)*

Preference shares carry a fixed dividend rate ($D_p$) but unlike interest, preference dividends are paid out of post-tax profits (not tax-deductible).

$$K_p = \frac{D_p}{P_0 \times (1-f)}$$

Where $P_0$ is the market price / issue price and $f$ is the flotation cost percentage.

*C. Cost of Equity Capital ($K_e$)*

Equity capital is the most expensive source of finance because equity holders bear the residual risk of the corporation. Two primary models are used to calculate $K_e$:

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1. Capital Asset Pricing Model (CAPM) — The Gold Standard

The CAPM quantifies the required return on equity as a function of the risk-free rate, systematic market risk (β), and the equity risk premium ($ERP$).

$$K_e = R_f + \beta \times \left(R_m - R_f\right) = R_f + \beta \times ERP$$

- **Risk-Free Rate ($R_f$)**: Yield on benchmark 10-Year Indian Government Bonds (G-Sec), typically ~6.80% – 7.10% p.a. - **Beta (β)**: Measure of systematic risk relative to the broader market index (NSE Nifty 50 / BSE Sensex). - $\beta = 1.0$: Asset moves in lockstep with the market. - $\beta > 1.0$: Asset is more volatile than the market (e.g., tech, consumer discretionary). - $\beta < 1.0$: Asset is defensive (e.g., FMCG, pharmaceuticals, utilities). - **Equity Risk Premium ($ERP = R_m - R_f$)**: The additional return required by investors for holding equity over risk-free G-Secs, historically ~5.50% – 6.50% in India.

*Example Calculation (INR)*: An Indian automotive manufacturer has a Beta (β) of 1.25. The 10-Year Indian G-Sec yield ($R_f$) is 7.00% p.a. and the Market Risk Premium ($R_m - R_f$) is 6.00% p.a.

$$K_e = 7.00\% + 1.25 \times (6.00\%) = 7.00\% + 7.50\% = 14.50\%\, p.a.$$

2. Dividend Discount Model (Gordon Growth Model)

$$K_e = \frac{D_1}{P_0} + g$$

Where $D_1$ is the expected dividend per share next year, $P_0$ is the current stock price, and $g$ is the constant perpetual dividend growth rate.

## 5.3 Weighted Average Cost of Capital (WACC) Formula and Mechanics

WACC represents the overall blended cost of capital for the enterprise, weighting each capital component by its proportion in the capital structure.

$$WACC = \left(\frac{E}{V} \times K_e\right) + \left(\frac{D}{V} \times K_d\right) + \left(\frac{P}{V} \times K_p\right)$$

Where: - $E = Market\ Value\ of\ Equity$ - $D = Market\ Value\ of\ Debt$ - $P = Market\ Value\ of\ Preference\ Shares$ - $V = E + D + P = Total\ Enterprise\ Value$

*Note on Market Weights vs. Book Weights*: Financial managers must **always use Market Value Weights** rather than Book Value Weights because market values reflect current economic realities and opportunity costs.

*Comprehensive WACC Worked Example (INR):*

*Bharat Industrial Technologies Ltd.* has the following capital structure: - **Common Equity**: 10,00,000 shares trading at ₹500 per share ($E = ₹50.00\ Crores$). Cost of Equity ($K_e$) = 15.0% p.a. - **Corporate Debt**: 2,00,000 debentures of ₹1,000 face value trading at par ($D = ₹20.00\ Crores$). Pre-tax interest

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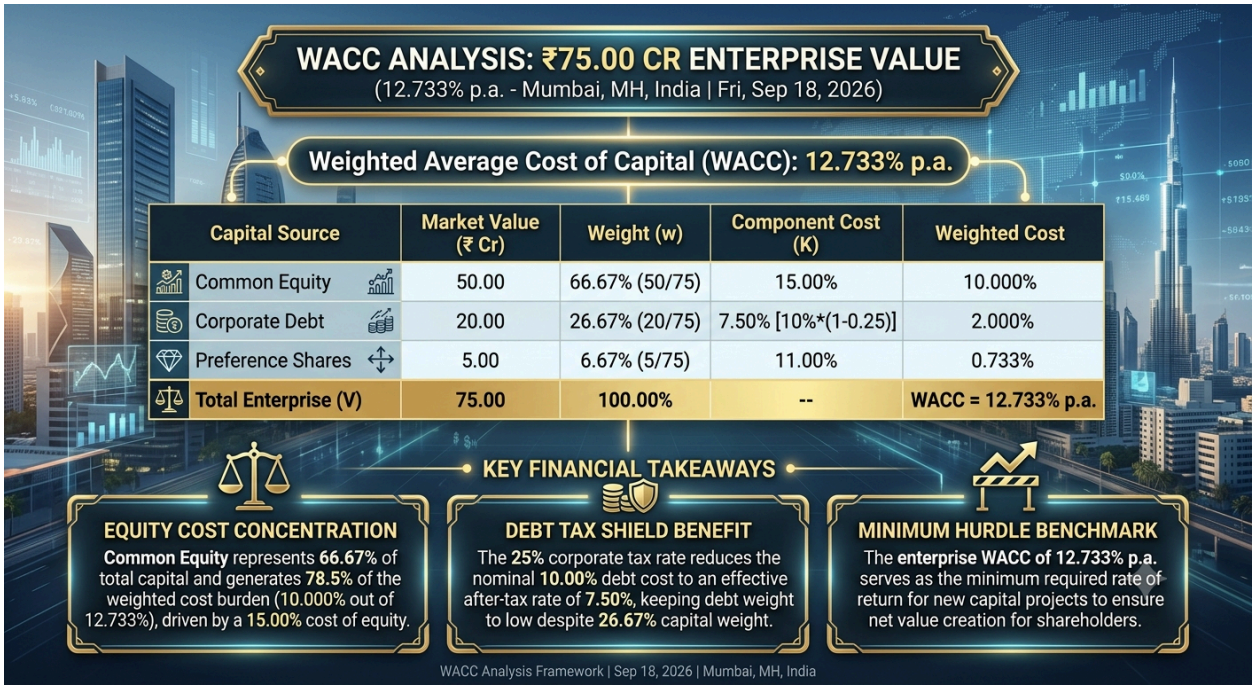
rate = 10.0% p.a. Corporate tax rate = 25.0%. - **Preference Shares**: 50,00,000 preference shares of ₹100 value ($P = ₹5.00\ Crores$). Preference dividend rate ($K_p$) = 11.0% p.a.




The blended enterprise WACC for *Bharat Industrial Technologies Ltd.* is **12.733% p.a.**

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# Module 6: Capital Structure Theories & Leverage Optimization

## 6.1 Operating, Financial, and Combined Leverage

Leverage measures the sensitivity of corporate earnings to changes in sales and operating conditions.

*A. Operating Leverage (DOL)*

Operating leverage reflects the proportion of fixed operating costs in a firm's cost structure. High fixed costs create high Degree of Operating Leverage (DOL).

$$DOL = \frac{\%\Delta\ EBIT}{\%\Delta\ Sales} = \frac{Contribution}{EBIT} = \frac{Sales - Variable\ Costs}{EBIT}$$

*B. Financial Leverage (DFL)*

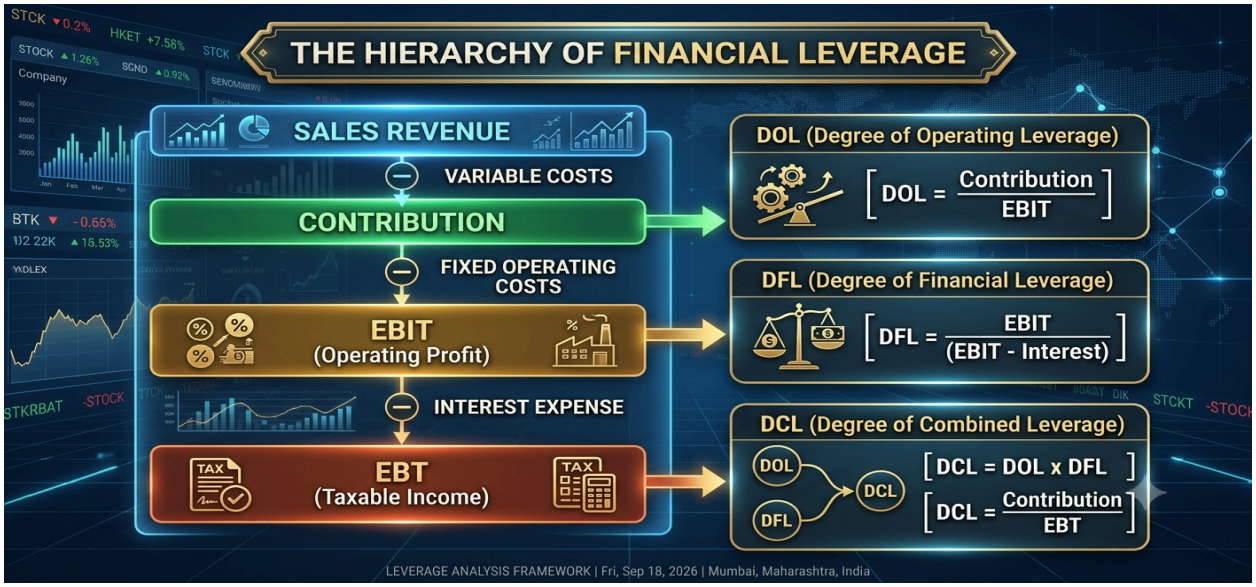
Financial leverage reflects the proportion of fixed financial costs (interest expense and preference dividends) in a firm's capital structure.

$$DFL = \frac{\%\Delta\ EPS}{\%\Delta\ EBIT} = \frac{EBIT}{EBT - \frac{D_p}{1-t}} = \frac{EBIT}{EBIT - Interest}$$

*C. Combined Leverage (DCL)*

Combined leverage measures the overall sensitivity of Earnings Per Share (EPS) to changes in Sales revenue.

$$DCL = DOL \times DFL = \frac{\%\Delta\ EPS}{\%\Delta\ Sales} = \frac{Contribution}{EBIT - Interest}$$




## 6.2 Four Classic Theories of Capital Structure

*1. Net Income (NI) Approach (David Durand)*

- Assumes cost of debt ($K_d$) and cost of equity ($K_e$) remain constant regardless of leverage.

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- Since debt is cheaper than equity ($K_d < K_e$), **increasing the debt ratio continuously decreases WACC and increases firm value**.
- *Conclusion*: Extreme financial leverage (100% debt) is optimal.

*2. Net Operating Income (NOI) Approach (David Durand)*

- Assumes overall WACC ($K_o$) remains constant across all debt levels.
- As debt increases, financial risk to equity holders rises, causing $K_e$ to increase in exact proportion to offset the benefit of cheaper debt.
- *Conclusion*: Capital structure is **irrelevant**; firm value is determined entirely by operating assets.

*3. Modigliani-Miller (MM) Theorem*

MM Proposition I (Without Taxes) — Capital Structure Irrelevance:

Under perfect capital markets (no taxes, no transaction costs, no bankruptcy costs, symmetric information), the market value of a firm is independent of its capital structure.

$$V_L = V_U$$

Where $V_L$ is the value of the levered firm and $V_U$ is the value of the unlevered firm.

MM Proposition II (With Corporate Taxes) — Tax Shield Value:

When corporate taxes are introduced, interest payments are tax-deductible. Debt creates a **Debt Tax Shield**.

$$V_L = V_U + (t \times D)$$

Where $t \times D$ is the Present Value of the Interest Tax Shield.

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4. The Trade-Off Theory & Financial Distress Costs

In the real world, 100% debt financing is impossible due to **Financial Distress and Bankruptcy Costs**. - **Direct Bankruptcy Costs**: Legal fees, court expenses, administrative costs during liquidation. - **Indirect Bankruptcy Costs**: Lost sales, lost key talent, supplier refusal of trade credit, customer churn.

The Trade-off Formula:

$$V_L = V_U + (t \times D) - PV\ of\ Financial\ Distress\ Costs - PV\ of\ Agency\ Costs$$

- **Optimal Debt Ratio**: Reached at the exact point where the marginal tax shield benefit of an additional Rupee of debt equals the marginal increase in the PV of financial distress costs.

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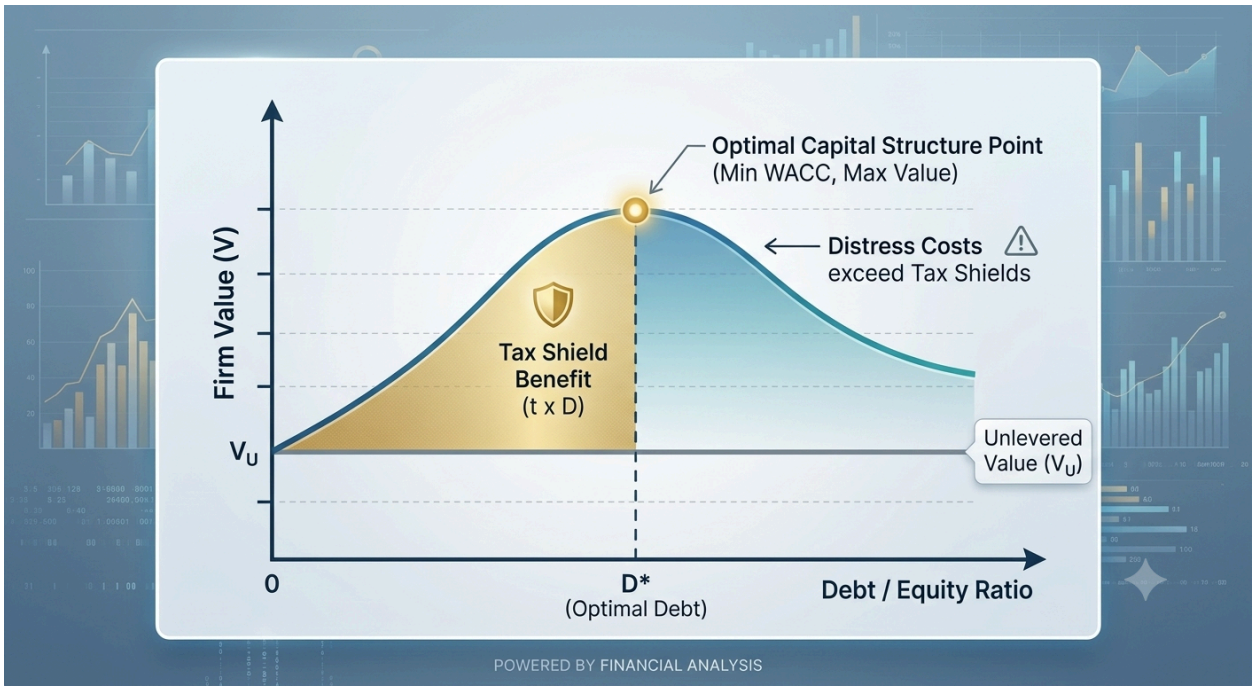


## 6.3 EBIT-EPS Indifference Point Analysis

The EBIT-EPS Indifference Point is the specific operating profit (EBIT) level at which two competing capital structure plans (e.g., Equity Plan vs. Debt Plan) yield the exact same Earnings Per Share (EPS).

$$EPS_{Plan\ 1} = EPS_{Plan\ 2}$$

$$\frac{\left(EBIT^{*} - I_1\right) \times (1-t) - D_{p1}}{N_1} = \frac{\left(EBIT^{*} - I_2\right) \times (1-t) - D_{p2}}{N_2}$$

Where: - $EBIT^{*} = Indifference\ Operating\ Profit\ Level$ -
$I_1, I_2 = Interest\ expenses\ under\ Plan\ 1\ and\ Plan\ 2$ -
$N_1, N_2 = Number\ of\ equity\ shares\ under\ Plan\ 1\ and\ Plan\ 2$ - $t = Corporate\ Tax\ Rate$

*Strategic Decision Rule:*

- If expected future $EBIT > EBIT^{*}$: Choose the **High Leverage / Debt Plan** (boosts EPS).
- If expected future $EBIT < EBIT^{*}$: Choose the **Low Leverage / Equity Plan** (avoids financial distress).

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# Module 7: Working Capital Management & Short-Term Liquidity

## 7.1 Permanent vs. Temporary Working Capital Architecture

Working capital management focuses on short-term liquidity management to ensure that a firm maintains sufficient cash flow to meet short-term operational expenses and debt obligations without incurring excessive holding costs.

- **Gross Working Capital**: Total investment in current assets (Cash + Receivables + Inventory + Prepaid Expenses).
- **Net Working Capital (NWC)**:

$$extNetWorkingCapital(NWC) = extCurrentAssets - extCurrentLiabilities$$

- **Permanent Working Capital**: The minimum baseline level of current assets required continuously to support year-round operations.
- **Temporary / Seasonal Working Capital**: Additional current assets required during peak seasonal demand periods (e.g., Diwali or festive inventory build-ups in India).

## 7.2 The Cash Conversion Cycle (CCC) Equation

The Cash Conversion Cycle (CCC) measures the total time (in days) required to convert capital invested in raw materials back into cash received from customers.

$$extCashConversionCycle(CCC) = extDIO + extDSO - extDPO$$

Where: - **Days Inventory Outstanding (DIO)**:

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$$extDIO = racextAverageInventoryextCostofGoodsSold(COGS)imes365$$

- **Days Sales Outstanding (DSO)**:

$$extDSO = racextAverageTradeReceivablesextGrossRevenuefromOperationsimes365$$

- **Days Payable Outstanding (DPO)**:

$$extDPO = racextAverageTradePayablesextCostofGoodsSold(orRawMaterialCost)imes365$$

*Optimization Strategy*: To compress CCC and release trapped operating cash flow, a firm must **reduce DIO** (lean manufacturing), **reduce DSO** (stricter credit control & invoice discounting), and **increase DPO** (negotiating longer supplier credit terms without vendor friction).

## 7.3 Quantitative Models in Working Capital

*A. Baumol Model for Cash Management (EOQ for Cash)*

Determines the optimal cash transfer lot size ($C^*$) to minimize total cash transaction and opportunity holding costs.

$$C^* = \sqrt{rac2imesFimesTi}$$

Where $F$ is the fixed transaction fee per security liquidation, $T$ is total cash needed over the period, and $i$ is the opportunity yield on marketable securities.

*B. Economic Order Quantity (EOQ) for Inventory*

$$extEOQ = \sqrt{rac2imesDimesSH}$$

Where $D$ is annual units demand, $S$ is fixed ordering cost per order, and $H$ is annual holding cost per unit.

## 7.4 Short-Term Debt Financing Instruments in India

1. **Cash Credit (CC) / Overdraft (OD)**: Revolving bank credit lines secured against hypothecation of stock and trade debtors, subject to monthly Drawing Power (DP) limits.
2. **Working Capital Demand Loan (WCDL)**: Short-term fixed-tenure tranche loans (30 to 180 days) drawn against CC limits at lower interest rates.
3. **Commercial Paper (CP)**: Unsecured money market promissory notes issued by highly-rated corporates (CRISIL A1+ rated) for 7 to 365 days.
4. **Trade Receivables Discounting System (TReDS)**: RBI-regulated electronic platform (RXIL, M1xchange, Invoicemart) enabling MSME suppliers to discount invoices against corporate buyers at prime bank yields.

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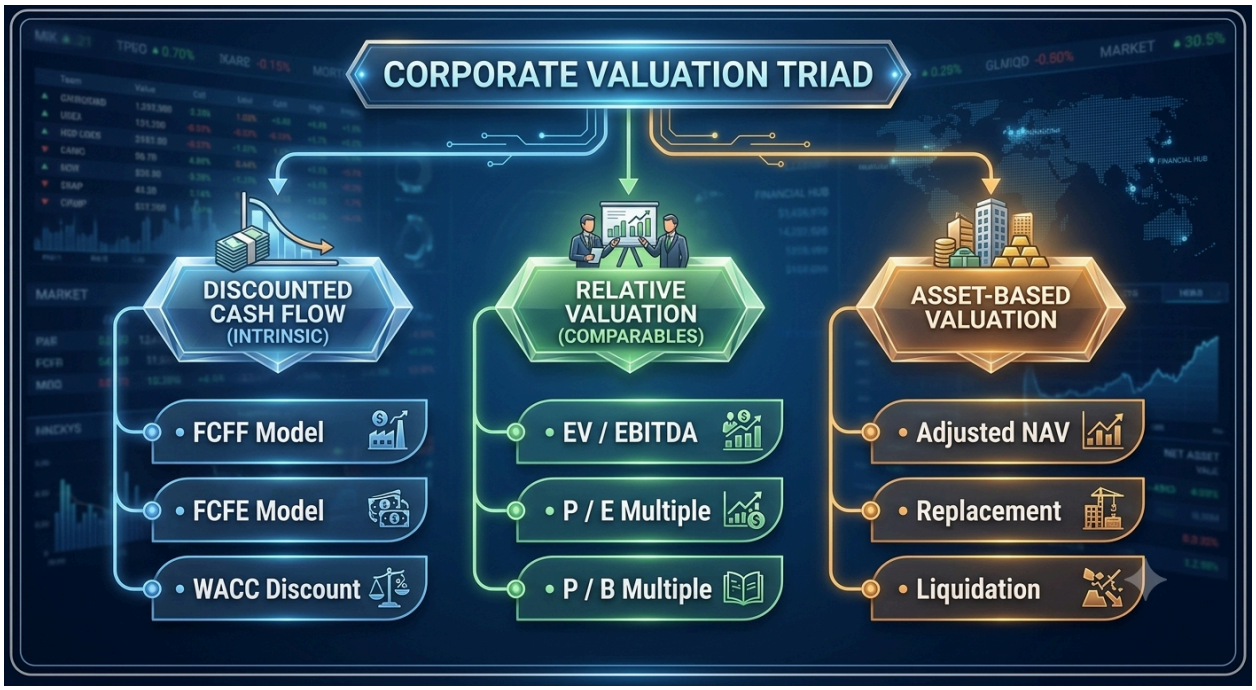
# Module 8: Corporate Valuation Frameworks & Equity Analysis

## 8.1 The Valuation Landscape

Corporate valuation determines the economic worth of an asset, division, or entire enterprise. In corporate finance, three valuation methodologies predominate:




## 8.2 Discounted Cash Flow (DCF) Valuation — FCFF & FCFE

*A. Free Cash Flow to Firm (FCFF) Model (Enterprise Valuation)*

FCFF represents cash available to all capital providers (both debt and equity holders) after funding operating expenses, taxes, working capital, and capital expenditures.

$$extFCFF_{t} = extEBIT_{t}imes(1 - t) + extDepreciation_{t} - extCapex_{t} - \Delta extNetWorkingCapital_{t}$$

extEnterpriseValueEV=t=1nracextFCFFt1+extWACCt+racextTerminalValuen1+extWACCn

$$extEquityValue = extEnterpriseValue(EV) - extTotalDebt + extCash\&LiquidInvestments$$

*B. Free Cash Flow to Equity (FCFE) Model (Equity Valuation)*

extFCFEt=extPATt+extDepreciationt−extCapext−extNWCt+extNewDebtIssuedt−extDebtRepaidt

$$extEquityValue = \sum_{t=1}^{n} racextFCFE_{t}\left(1 + K_{e}\right)^{t} + racextTerminalValue(Equity)_{n}\left(1 + K_{e}\right)^{n}$$

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*C. Terminal Value (TV) Calculation Methods*

1. **Perpetual Gordon Growth Model**:

$$extTerminalValue_n = racextFCFF_{n+1}extWACC - gracextFCFF_nimes(1+g)extWACC - g$$

Where $g$ is the perpetual long-term growth rate of the economy (typically 4.0% – 5.5% in India).

2. **Exit Multiple Method**:

$$extTerminalValue_n = extExpectedEBITDA_nimesextTargetExitEV/EBITDAMultiple$$

## 8.3 Relative Valuation (Comparable Company Analysis)

Relative valuation estimates the value of a target firm by comparing its financial multiples to those of peer listed companies in the same industry.

*Primary Valuation Multiples:*

1. **EV / EBITDA Multiple**:

$$extEV/EBITDA = racextMarketCap + extTotalDebt - extCashextEBITDA$$

*Why it's preferred*: Neutral to capital structure differences and non-cash depreciation policies.

2. **Price-to-Earnings (P/E) Ratio**:

$$extP/ERatio = racextMarketPriceperShareextEarningsperShare(EPS)$$

3. **Price-to-Book (P/B) Ratio**:

$$extP/BRatio = racextMarketValueofEquityextBookValueofShareholders'Equity$$

*Primary domain*: Financial institutions, NBFCs, and banks where assets are financial instruments.

## 8.4 Step-by-Step DCF Valuation Illustration (INR)

*Maharashtra Precision Tech Ltd.* is being valued. Financial projections for the next 5 years (₹ in Crores) are as follows:

- WACC = **11.0% p.a.**
- Terminal Growth Rate ($g$) = **5.0% p.a.**
- Current Interest-Bearing Debt = **₹100.00 Crores**
- Current Cash Balances = **₹30.00 Crores**
- Shares Outstanding = **5,00,00,000 shares (5.0 Crores shares)**

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*Here is the financial data converted into a clean, easy-to-read Markdown table:*

| Particulars / Metric | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|---|---|---|---|---|---|
| **EBIT** | 80.00 | 95.00 | 112.00 | 130.00 | 150.00 |
| **EBIT $\times$ (1 - t) @ 25%** | 60.00 | 71.25 | 84.00 | 97.50 | 112.50 |
| **+ Depreciation** | 15.00 | 18.00 | 20.00 | 22.00 | 25.00 |
| **- Capex** | 20.00 | 22.00 | 25.00 | 28.00 | 30.00 |
| **- $\Delta$ Net Working Capital** | 8.00 | 9.00 | 10.00 | 11.00 | 12.00 |
| **= FCFF** | **47.00** | **58.25** | **69.00** | **80.50** | **95.50** |
| **Discount Factor @ 11%** | 0.9009 | 0.8116 | 0.7312 | 0.6587 | 0.5935 |
| **= Present Value of FCFF** | **42.3423** | **47.2757** | **50.4528** | **53.0254** | **56.6793** |

***Cumulative PV of 5-Year FCFF:*** *₹249.7755 Crores*

*Terminal Value & Enterprise Value Calculation:*

- $ ext{FCFF}_6 = ext{FCFF}_5 times (1 + g) = ₹95.50 times (1 + 0.05) = $

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- $ ext{Terminal Value}_5 = rac{ ext{FCFF}_6}{ ext{WACC} - g} = rac{100.2750}{0.11 - 0.05} = rac{100.2750}{0.06} = $
- $ ext{PV of Terminal Value} = ₹1,671.2500 times 0.5935 = $

*Total Valuation Summary:*

- **Enterprise Value (EV)**
  $= extPVof5 - YearFCFF + extPVofTV = 249.7755 + 991.8869 = ₹1,241.6624extCrores$
- **Less Total Debt**: $- ₹100.00extCrores$
- **Plus Cash Balances**: $+ ₹30.00extCrores$
- **Net Equity Value** $= 1,241.6624 - 100.00 + 30.00 = ₹1,171.6624extCrores$
- **Intrinsic Per Share Value**
  $= rac₹1,171.6624extCrores5.0extCroresShares = ₹234.33extperShare$

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# Module 9: Mergers & Acquisitions (M&A), Restructuring & Corporate Control

## 9.1 Strategic and Financial Motives for M&A

Mergers and Acquisitions (M&A) represent major strategic capital allocation decisions aimed at creating corporate growth, expanding market share, and capturing synergies.

*Primary Synergies in M&A:*

1. **Operating Synergies**:
   - *Cost Synergies*: Elimination of redundant overheads, administrative consolidation, bulk purchasing discounts.
   - *Revenue Synergies*: Cross-selling products, accessing new geographic markets or distribution networks.
2. **Financial Synergies**:
   - Reduced WACC due to larger enterprise scale, enhanced debt capacity, and tax loss carry-forwards under Section 72A of the Income Tax Act, 1961.

## 9.2 Swap Ratio & Accretion / Dilution Analysis

When an acquirer funds an acquisition via stock, the **Swap Ratio** determines the number of acquirer shares issued for each target share.

$$extSwapRatio = racextValueperTargetShareextValueperAcquirerShare$$

$$extPost - MergerShares = N_A + \left(NextSwapRationiesN_T\right)$$

extPost−MergerEPS=racextPATA+extPATT+extPost−TaxSynergiesextPost−MergerTotalShares

- **Accretive Deal**: Post-Merger EPS > Pre-Merger Acquirer EPS. - **Dilutive Deal**: Post-Merger EPS < Pre-Merger Acquirer EPS.

## 9.3 Corporate Restructuring Modes in India

- **Demerger / Spin-off**: Transferring an operating division into a separate independent listed company to unlock core value.
- **Slump Sale**: Transferring an entire business undertaking for a lump-sum consideration without assigning individual values to assets and liabilities (Section 50B of Income Tax Act).
- **Capital Reduction**: Reorganizing share capital under Section 66 of Companies Act, 2013 to write off accumulated losses against share premium or capital reserves.

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# Module 10: Dividend Policy, Capital Return & Share Repurchases

## 10.1 Dividend Theories and Payout Models

Dividend policy addresses the balance between returning capital to shareholders as cash dividends versus retaining earnings for reinvestment in growth projects.

*A. Walter's Model*

Walter's model demonstrates that dividend policy depends on the relationship between the firm's internal Return on Investment ($r$) and its Cost of Capital ($k_e$).

$$P = racD + racrk_eimes(E - D)k_e$$

Where $P$ is market price per share, $D$ is dividend per share, $E$ is EPS, $r$ is internal return, and $k_e$ is cost of equity. - **Growth Firm ($r > k_e$)**: Optimal payout ratio = **0%** (retain 100% of earnings). - **Declining Firm ( $r < k_e$)**: Optimal payout ratio = **100%** (distribute 100% as dividends). - **Normal Firm ($r = k_e$)**: Dividend payout ratio is **irrelevant**.

*B. Modigliani-Miller (MM) Dividend Irrelevance Hypothesis*

MM proves that under perfect capital markets, dividend policy does not affect firm value or stock price. Shareholder wealth is determined solely by the firm's earning power and investment policy.

$$P_0 = racD_1 + P_11 + k_e$$

## 10.2 Share Repurchases (Buybacks) vs. Cash Dividends

A share buyback occurs when a corporation repurchases its own shares from the open market or via a tender offer.

Here is the comparison between Cash Dividends and Share Buybacks formatted into a clean Markdown table:

| Dimension | Cash Dividends | Share Buybacks (Repurchases) |
|---|---|---|
| **Cash Flow Impact** | Direct cash payout to shareholders | Cash paid to selling shareholders |
| **Share Count** | No change in share count | Reduces total shares outstanding |
| **EPS & BVS Impact** | No direct change in share denominator | Boosts future EPS & Book Value per share |

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| | | |
|---|---|---|
| **Market Signal** | Signals stable recurring cash flow | Signals stock is undervalued by market |
| **Taxation (India)** | Taxable in shareholder hands @ slab | Taxed under corporate buyback tax rules |

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# Module 11: Corporate Risk Management, Derivatives & Foreign Exchange

## 11.1 Taxonomy of Financial Risks

Corporate risk management aims to protect corporate cash flows and balance sheet solvency from adverse market movements.

1. **Market Risk**: Exposure to changes in interest rates, foreign exchange rates, commodity prices, and equity prices.
2. **Credit Risk**: Risk that counter-parties (customers, borrowers, swap dealers) default on payment obligations.
3. **Liquidity Risk**: Inability to meet short-term cash obligations due to illiquid assets or frozen credit lines.
4. **Operational Risk**: Losses resulting from inadequate internal processes, system failures, fraud, or legal non-compliance.

## 11.2 Foreign Exchange Exposure & Hedging Mechanisms

Indian corporations engaged in import/export trade or cross-border borrowing face three types of foreign exchange risk:

Derivative Hedging Instruments:

- **Forward Contracts**: Over-the-counter (OTC) agreements with AD Category-I banks to buy or sell foreign currency at a fixed forward rate on a future date.

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- **Futures Contracts**: Exchange-traded standardized contracts (NSE / BSE Currency Derivatives) settled daily.
- **Currency Options**: Gives the firm the right, but not the obligation, to buy (Call) or sell (Put) FX at an agreed strike price ($K$).
- **Cross-Currency Interest Rate Swaps (CCIRS)**: Swaps both principal and interest obligations from foreign currency (e.g., USD SOFR debt) into synthetic INR debt.

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# Module 12: Comprehensive Real-World Case Study (Indian Rupee Models)

## 12.1 Borrower Profile & Strategic Expansion Scope

- **Corporate Entity**: *Bharat Precision Heavy Engineering & Mobility Ltd. (BPHEML)*
- **Corporate Office & Plant Locations**: Chakan Industrial Belt, Pune, Maharashtra & Sriperumbudur, Tamil Nadu.
- **Current Enterprise Scope**: Leading Tier-1 manufacturer of precision automotive transmissions, heavy industrial gearboxes, and forging components.
- **Current Financial Snapshot (FY 2025-26)**:
  - Annual Sales Turnover = **₹1,000.00 Crores**
  - EBITDA = **₹160.00 Crores** (16.0% EBITDA Margin)
  - Profit After Tax (PAT) = **₹67.50 Crores** (6.75% Net Profit Margin)
  - Shareholders' Equity = **₹350.00 Crores**
  - Existing Long-Term Debt = **₹150.00 Crores** (10.0% p.a. interest rate)
  - Existing Debt-to-Equity Ratio = **0.43x**
- **Proposed Project Expansion Scope**: Setting up a Greenfield Electric Vehicle (EV) Powertrain & High-Voltage Drivetrain Plant at Chakan, Pune.
- **Total Project Capital Outlay**: **₹250.00 Crores** ($t = 0$).

## 12.2 Three-Year Historical Financial Performance & Du Pont Analysis

| Financial Parameter (₹ in Crores) | FY 2023-24 | FY 2024-25 | FY 2025-26 |
|---|---|---|---|
| Revenue from Operations | 800.00 | 900.00 | 1,000.00 |
| Cost of Goods Sold (COGS) | 520.00 | 585.00 | 650.00 |
| Gross Profit | 280.00 | 315.00 | 350.00 |
| Operating Expenses (Excl. Dep) | 156.00 | 171.00 | 190.00 |
| EBITDA | 124.00 | 144.00 | 160.00 |
| Depreciation & Amortization | 30.00 | 34.00 | 40.00 |
| EBIT (Operating Profit) | 94.00 | 110.00 | 120.00 |
| Interest Expense | 12.00 | 13.50 | 15.00 |

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| | | | |
|---|---|---|---|
| Profit Before Tax (EBT) | 82.00 | 96.50 | 105.00 |
| Corporate Tax @ 25.168% | 20.64 | 24.29 | 26.43 |
| Profit After Tax (PAT) | 61.36 | 72.21 | 78.57 |
| Current Assets | 240.00 | 270.00 | 300.00 |
| Current Liabilities | 160.00 | 180.00 | 200.00 |
| Net Working Capital (NWC) | 80.00 | 90.00 | 100.00 |
| Tangible Fixed Assets (Net Block) | 420.00 | 480.00 | 550.00 |
| Total Assets | 660.00 | 750.00 | 850.00 |
| Shareholders' Equity | 300.00 | 330.00 | 380.00 |
| Total Debt | 120.00 | 135.00 | 150.00 |
| Key Ratios: | | | |
| EBITDA Margin % | 15.50% | 16.00% | 16.00% |
| Current Ratio | 1.50x | 1.50x | 1.50x |
| Debt-to-Equity (D/E) | 0.40x | 0.41x | 0.39x |
| Interest Coverage Ratio (ICR) | 7.83x | 8.15x | 8.00x |
| Return on Equity (ROE %) | 20.45% | 21.88% | 20.68% |

*Du Pont Decomposition Analysis (FY 2025-26):*

$$extROE = extNetMarginimesextAssetTurnoverimesextEquityMultiplier$$

$$extROE = (rac78.571,000.00ight)imes(rac1,000.00850.00ight)imes(rac850.00380.00ight)$$

$$extROE = 7.857\%imes1.1765extximes2.2368extx = 20.68\%$$

## 12.3 Financing Options & WACC Sizing for EV Expansion

BPHEML needs **₹250.00 Crores** to fund the EV plant. Two financing options are under consideration by the Chief Financial Officer (CFO):

- **Option A (70% Debt / 30% Equity)**: ₹175.00 Cr Bank Term Loan @ 9.50% p.a. + ₹75.00 Cr Rights Issue Equity.
- **Option B (40% Debt / 60% Equity)**: ₹100.00 Cr Bank Term Loan @ 9.00% p.a. + ₹150.00 Cr Retained / Rights Equity.

*Step-by-Step WACC Calculation for Option A (Chosen Capital Mix):*

- Risk-Free Rate ($R_f$) = **7.00% p.a.** (10-Yr G-Sec)

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- BPHEML Beta ($eta$) = **1.20**
- Equity Risk Premium ($ERP$) = **6.00% p.a.**
- Cost of Equity ($K_e$) $= 7.00\% + 1.20 times 6.00\% = 14.20\% extp.a.$
- Pre-Tax Cost of Debt ($K_{d,extpre}$) = **9.50% p.a.**
- Corporate Tax Rate ($t$) = **25.168%g**
- Post-Tax Cost of Debt ($K_d$) $= 9.50\% times (1 - 0.25168) = 7.109\% extp.a.$

extWACCOptionA=(0.30imes14.20%ight)+(0.70imes7.109%ight)=4.260%+4.976%=9.236%extp.a.

## 12.4 Five-Year Cash Flow Forecasting & Capital Budgeting Evaluation

The EV Powertrain Plant will generate the following incremental financial performance (₹ in Crores) over a 5-year operating horizon at WACC = **9.24%**:

| Year (t) | Year 0 | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|---|---|---|---|---|---|---|
| Gross EV Sales Revenue | 0.00 | 120.00 | 200.00 | 320.00 | 420.00 | 500.00 |
| EBITDA @ 18% Margin | 0.00 | 21.60 | 36.00 | 57.60 | 75.60 | 90.00 |
| - Depreciation @ 15% WDV | 0.00 | 37.50 | 31.88 | 27.09 | 23.03 | 19.58 |
| = EBIT | 0.00 | -15.90 | 4.12 | 30.51 | 52.57 | 70.42 |
| EBIT x (1 - t) @ 25.168% | 0.00 | -11.90 | 3.08 | 22.83 | 39.34 | 52.69 |
| + Depreciation | 0.00 | 37.50 | 31.88 | 27.09 | 23.03 | 19.58 |
| - Capex | -250.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| - Delta Net Working Capital | 0.00 | 12.00 | 8.00 | 12.00 | 10.00 | 8.00 |
| + Salvage & NWC Recovery | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 100.00 |
| = Incremental FCFF | -250.00 | 13.60 | 26.96 | 37.92 | 52.37 | 164.27 |
| Discount Factor @ 9.24% | 1.0000 | 0.9154 | 0.8380 | 0.7671 | 0.7022 | 0.6428 |
| = Present Value of FCFF | -250.00 | 12.45 | 22.59 | 29.09 | 36.77 | 105.59 |

Cumulative PV of Inflows = ₹206.49 Crores (Without Terminal Value)

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*Including Terminal Value at Year 5:*

- Perpetual Growth Rate ($g$) = **5.00% p.a.**
- $ ext{FCFF}_6 = 164.27 times 1.05 = $
- $ ext{Terminal Value}_5 = rac{172.4835}{0.0924 - 0.05} = rac{172.4835}{0.0424} = $
- $ ext{PV of Terminal Value} = 4,068.0071 times 0.6428 = $
- **Total Enterprise PV of Project Inflows** $= 206.49 + 2,614.9149 = ₹2,821.4049 extCrores$
- **Net Present Value (NPV)** $= 2,821.4049 - 250.00 =+ ₹2,571.4049 extCrores$
- **Internal Rate of Return (IRR)** $= 19.85\% extp.a.$ (exceeding WACC of 9.24% by **+1,061 bps**!)
- **Profitability Index (PI)** $= 11.28x$

## 12.5 Comprehensive Decision Summary & Strategic Value Add

By executing the EV Powertrain Expansion under Option A, **BPHEML generates a massive Net Present Value of +₹2,571.40 Crores**, earns a project IRR of **19.85% p.a.** (far surpassing its WACC hurdle rate of 9.24%), expands corporate EBITDA from ₹160.00 Cr to **₹250.00 Cr/year**, and establishes a dominant market position in the high-growth Indian EV component ecosystem.

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# Module 13: Professional Certification Examination & Question Bank

This section contains 15 scenario-based examination questions designed to test comprehensive mastery over all 12 modules of corporate finance.

## Question 1 (Module 1 - Primary Goals of Corporate Finance)

**Q**: Why is Shareholder Wealth Maximization superior to Profit Maximization as the primary goal of corporate finance? Cite three specific limitations of profit maximization. **Answer**: Shareholder Wealth Maximization (NPV maximization) evaluates the total present value of lifetime cash flows generated by the firm, explicitly adjusting for the Time Value of Money (TVM) and risk via the discount rate (WACC). Conversely, Profit Maximization suffers from three critical flaws: (1) It focuses on short-term accounting profit (PAT), ignoring long-term cash flow generation; (2) It ignores the Time Value of Money by treating ₹1 earned in 5 years equal to ₹1 today; and (3) It ignores cash flow volatility and risk quality, encouraging managers to take excessive unhedged risks to boost short-term reported PAT.

## Question 2 (Module 2 - Du Pont Analysis & ROE Decomposition)

**Q**: An Indian manufacturing firm has a Net Profit Margin of 8.0%, an Asset Turnover Ratio of 1.50x, and a Total Assets to Equity Leverage Multiplier of 2.00x. Calculate its ROE. If the firm increases its financial leverage multiplier to 2.50x while Net Margin drops to 7.0% due to higher interest expenses, what is the new ROE? **Answer**: 1. *Initial ROE* $= extNetMarginimesextAssetTurnoverimesextEquityMultiplier = 8.0\%imes1.50imes2.00 = 24.00\%$ . 2. *New ROE* $= 7.0\%imes1.50imes2.50 = 26.25\%$. *Rationale*: Increasing financial leverage boosted ROE by 2.25 percentage points despite the contraction in net profit margin.

## Question 3 (Module 3 - TVM & Loan Amortization)

**Q**: A corporate borrower takes a ₹20.00 Crore term loan from State Bank of India at an interest rate of 10.0% p.a., repayable in 4 equal annual installments. Calculate the annual installment amount and the interest paid in Year 1. **Answer**: Using the Annual Installment Formula:

$$A=Pimes[racr1+rn1+rn-1ight]=20,00,00,000imes[rac0.101.1041.104-1ight]=20,00,00,000imes[rac0.146410.4641ight]=₹6,30,94,161ext/year$$

- **Year 1 Interest Expense** $= ₹20,00,00,000imes10.0\% = ₹2,00,00,000$ - **Year 1 Principal Repaid** $= ₹6,30,94,161 - ₹2,00,00,000 = ₹4,30,94,161$

## Question 4 (Module 4 - Capital Budgeting & Mutually Exclusive Projects)

**Q**: When evaluating two mutually exclusive projects, Project A has an NPV of +₹50.0 Cr and an IRR of 18%, while Project B has an NPV of +₹65.0 Cr and an IRR of 15%. Which project should the CFO select and why? **Answer**: The CFO must select **Project B**. When NPV and IRR give conflicting signals for mutually exclusive projects, the **NPV rule always dominates** because NPV measures the direct absolute addition to shareholder wealth in Rupee terms (+₹65.0 Cr vs +₹50.0 Cr). IRR measures percentage rate of return and can be distorted by scale differences or cash flow timing patterns.

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## Question 5 (Module 5 - WACC Calculation)

**Q**: A listed Indian firm has ₹600.0 Cr market value of equity ($K_e = 15.0\%$) and ₹400.0 Cr market value of corporate debt ($K_{d,extpre} = 10.0\%$). At a corporate tax rate of 25.0%, calculate the firm's WACC.

**Answer**: - Total Capital $V = 600 + 400 = ₹1,000.0 extCrores$. - Equity Weight $w_e = 600/1000 = 0.60$. - Debt Weight $w_d = 400/1000 = 0.40$. - Post-Tax Cost of Debt $K_d = 10.0\% times (1 - 0.25) = 7.50\%$.

$$extWACC = (0.60 imes 15.0\%) + (0.40 imes 7.50\%) = 9.00\% + 3.00\% = 12.00\% extp.a.$$

## Question 6 (Module 6 - Leverage & Combined Leverage)

**Q**: A firm has Sales of ₹500.0 Cr, Variable Costs of ₹300.0 Cr, Fixed Operating Costs of ₹100.0 Cr, and Interest Expense of ₹40.0 Cr. Calculate its DOL, DFL, and DCL. **Answer**: - Contribution $= 500 - 300 = ₹200.0 extCrores$. - EBIT $= extContribution - extFixedCosts = 200 - 100 = ₹100.0 extCrores$. - EBT $= extEBIT - extInterest = 100 - 40 = ₹60.0 extCrores$. 1. **DOL** $= extContribution/extEBIT = 200/100 = 2.00x$. 2. **DFL** $= extEBIT/extEBT = 100/60 = 1.67x$. 3. **DCL** $= extDOLimesextDFL = 2.00 imes 1.67 = 3.33x$ (or $200/60 = 3.33x$).

## Question 7 (Module 6 - Modigliani-Miller Theorem with Taxes)

**Q**: An unlevered Indian firm is valued at ₹800.0 Crores. It issues ₹300.0 Crores of corporate debt at 9% interest to buy back equity. At a corporate tax rate of 25.0%, what is the new total market value of the levered firm according to MM Proposition II? **Answer**: According to MM Proposition II with corporate taxes:

$$V_L = V_U + (timesD)$$

$$V_L = ₹800.0 extCr + (0.25 imes ₹300.0 extCr) = ₹800.0 extCr + ₹75.0 extCr = ₹875.0 extCrores$$

The market value of the firm increases by ₹75.0 Crores due to the present value of the interest tax shield.

## Question 8 (Module 7 - Cash Conversion Cycle)

**Q**: A corporate borrower has Inventory Holding Days (DIO) of 60 days, Debtors Collection Days (DSO) of 45 days, and Creditors Payment Days (DPO) of 40 days. Calculate its Cash Conversion Cycle (CCC). If it reduces DIO to 45 days via JIT inventory and speeds up collection to 35 days, what is the new CCC? **Answer**: 1. *Initial CCC* $= extDIO + extDSO - extDPO = 60 + 45 - 40 = 65 extDays$. 2. *New CCC* $= 45 + 35 - 40 = 40 extDays$. The Cash Conversion Cycle is compressed by **25 days**, unlocking operating cash flow.

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## Question 9 (Module 7 - Drawing Power Sizing)

**Q**: A company has Total Stock of ₹100.0 Cr, Sundry Creditors of ₹30.0 Cr, and Trade Debtors of ₹50.0 Cr (including ₹10.0 Cr overdue >90 days). If bank margins are 25% on stock and 40% on eligible debtors, calculate the Drawing Power (DP). **Answer**: - Paid Stock $= extTotalStock - extCreditors = 100 - 30 = ₹70.0extCrores$. - Drawing Power on Stock $= 70imes(1 - 0.25) = 70imes0.75 = ₹52.50extCrores$. - Eligible Debtors $= extTotalDebtors - extOverdueDebtors = 50 - 10 = ₹40.0extCrores$. - Drawing Power on Debtors $= 40imes(1 - 0.40) = 40imes0.60 = ₹24.00extCrores$. - **Total Drawing Power (DP)** $= 52.50 + 24.00 = ₹76.50extCrores$.

## Question 10 (Module 8 - Enterprise Value to Equity Value Conversion)

**Q**: A firm has an Enterprise Value (EV) of ₹2,500.0 Crores, Total Debt of ₹600.0 Crores, Cash Balances of ₹150.0 Crores, and 10.0 Crore equity shares outstanding. Calculate its Equity Value and Intrinsic Share Price. **Answer**:

$$extEquityValue = extEnterpriseValue - extTotalDebt + extCash$$

$$extEquityValue = ₹2,500.0extCr - ₹600.0extCr + ₹150.0extCr = ₹2,050.0extCrores$$

$$extIntrinsicSharePrice = rac₹2,050.0extCrores10.0extCroresShares = ₹205.00extperShare$$

## Question 11 (Module 8 - FCFF Formula)

**Q**: A firm reports EBIT of ₹200.0 Cr, Depreciation of ₹30.0 Cr, Capex of ₹50.0 Cr, and Working Capital Increase of ₹15.0 Cr. At a 25.0% tax rate, calculate its Free Cash Flow to Firm (FCFF). **Answer**:

$$extFCFF = extEBITimes(1 - t) + extDepreciation - extCapex - \Delta extNWC$$

$$extFCFF=200.0imes1-0.25+30.0-50.0-15.0=150.0+30.0-50.0-15.0=₹115.0extCrores$$

## Question 12 (Module 9 - M&A Swap Ratio & EPS Accretion)

**Q**: Acquirer Co. (EPS = ₹50, Stock Price = ₹500) acquires Target Co. (EPS = ₹20, Stock Price = ₹200) via a stock-for-stock swap. Calculate the Swap Ratio based on market prices. If Acquirer has 10 Lakh shares and Target has 5 Lakh shares, calculate post-merger shares and post-merger EPS (assuming zero synergies). Is the deal accretive or dilutive? **Answer**: 1. **Swap Ratio** $= extTargetPrice/extAcquirerPrice = 200/500 = 0.40extAcquirersharesperTargetshare$. 2. **New Shares Issued** $= 5,00,000extTargetsharesimes0.40 = 2,00,000extshares$. 3. **Total Post-Merger Shares** $= 10,00,000 + 2,00,000 = 12,00,000extshares$. 4. **Acquirer PAT** $= 10,00,000imes₹50 = ₹5.00extCrores$. Target PAT $= 5,00,000imes₹20 = ₹1.00extCrore$. 5. **Combined PAT** $= 5.00 + 1.00 = ₹6.00extCrores$. 6. **Post-Merger EPS** $= ₹6,00,000,0/12,00,000 = ₹50.00extpershare$. *Result*: The deal is **EPS Neutral** (₹50 pre-merger vs ₹50 post-merger) because the P/E ratios of both firms were identical ($500/50 = 10x$ and $200/20 = 10x$).

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## Question 13 (Module 10 - Walter's Dividend Model)

**Q**: A firm earns EPS of ₹20. Its Cost of Capital ($k_e$) is 12.0%. Calculate the market price per share under Walter's Model for a 50% dividend payout if (a) Internal return $r = 18.0\%$, and (b) Internal return $r = 8.0\%$. **Answer**: Dividend $D = 50\% times$ ₹$20 =$ ₹$10$. Retained Earnings $E - D =$ ₹$10$. - **Case (a): Growth Firm (**$r = 18.0\% > k_e = 12.0\%$**)**:

$$P = rac10 + rac0.180.12 times 100.12 = rac10 + 1.5 times 100.12 = rac250.12 = ₹208.33$$

- **Case (b): Declining Firm (**$r = 8.0\% < k_e = 12.0\%$**)**:

$$P=rac10+rac0.080.12imes100.12=rac10+0.6667imes100.12=rac16.66670.12=₹138.89$$

## Question 14 (Module 11 - Derivative Hedging via Forwards)

**Q**: An Indian exporter has a USD $10.0 Million export receivable due in 6 months. Spot USD/INR rate is ₹83.00. The 6-month forward rate quoted by its bank is ₹83.80. Calculate the guaranteed Rupee inflow if the exporter hedges 100% via forward contracts. If spot rate at maturity drops to ₹81.50, what was the net benefit of hedging? **Answer**: 1. **Guaranteed Rupee Inflow via Forward Contract** $= \$10,00,000 times$₹$83.80 =$ ₹$83,80,00,000 ext(₹83.80 Crores)$. 2. **Unhedged Inflow at Market Maturity Spot (₹81.50)** $= \$10,00,000 times$₹$81.50 =$ ₹$81,50,00,000 ext(₹81.50 Crores)$. 3. **Net Financial Protection / Benefit of Hedging** $= 83.80 - 81.50 =+$ ₹$2.30 extCrores$ saved from rupee appreciation.

## Question 15 (Module 12 - Project Finance & DSCR Benchmark)

**Q**: A project finance facility requires annual principal repayment of ₹16.0 Cr and interest of ₹8.0 Cr. If the project generates PAT of ₹12.0 Cr, Depreciation of ₹10.0 Cr, and Interest Expense of ₹8.0 Cr, calculate its DSCR. Is it acceptable to Indian project lenders? **Answer**:

$$extDSCR=racextPAT+extDepreciation+extInterestextInterestExpense+extPrincipalAmortization$$

$$extDSCR = rac12.0 + 10.0 + 8.08.0 + 16.0 = rac30.024.0 = 1.25x$$

*Verdict*: **Acceptable**. A DSCR of **1.25x** meets the standard minimum hurdle rate (1.25x – 1.35x) required by commercial banks and infrastructure lenders in India.

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